

Abstract Algebra

: Lecture 9

Leo

2024.10.22

Theorem 1. A_n is a simple group when $n \geq 5$.

Remark 2. Let $G = \text{Sym}(\Omega) = S_n$, if $|\Omega| = n$.

- each element of G is a product of transpositions. ✓.
- each even permutation of Ω is a product of some 3-cycles.

证明. Let $N \triangleleft A_n$, s.t. $N \neq \{e\}$. We aim to prove $N = A_n$.

Claim 1: N contains a 3-cycle.

Let $e \neq g \in N$ s.t. $|\text{Fix}(g)| \geq |\text{Fix}(x)|$ for all $x \in A_n$. In which $\text{Fix}(x) = \{\omega \in \Omega \mid \omega^x = \omega, x \in S_n\}$ called the set of fixed points of x . 拥有最多不动点

Remark 3. Since g is an even permutation, $|\text{Fix}(g)| \leq n - 3$.

If $|\text{Fix}(g)| = n - 3$ then g is a 3-cycle as claimed.

To complete the proof of the claim, we assume that $|\text{Fix}(g)| < n - 3$. Then relabelling if necessary,

we may write $g = \begin{cases} (12345 \dots) \dots \\ (1234)(5 \dots) \dots \\ (123)(45 \dots) \dots \\ (12)(34)(5 \dots) \dots \end{cases}$ or $g = (12)(34)$ which means that we write g into a product of disjoint cycles and put the longest cycle in the first place, and relabelling if necessary. g 1 g.

If $\text{Fix}(g) \cap \{1, 2, 3, 4, 5\} = \emptyset$ and so $\text{Fix}(g) \subseteq \{6, 7, \dots, n\}$. In particular, $1^g = 2$.

Let $\sigma = (345)$ and $h = [\sigma, g]$. Then $h \in N$ and $1^\sigma = 2^{\sigma^{-1}} = 1$. So $1 \in \text{Fix}(h)$ and h fixes each point in $\text{Fix}(g)$. Thus $|\text{Fix}(h)| \geq |\text{Fix}(g)| + 1$, and $h = 1$ by our assumption. Then $[\sigma, g] = 1 \Rightarrow \sigma g = g \sigma$ it is obviously wrong, just need to consider the result of conjugacy action of σ . sigma g sigma^{-1}

This is a contradiction, so g is not a product of two disjoint cycles. It is just a 3-cycle as we claimed. (or: contra. with g has the most # of fixed points in G.)

Remark 4. If $g = (12)(34)$ then $h = (354)$ also has more fixed points.

Lemma 5. Let $g = (ijk \dots)$, $\sigma \in S_n$, $g^\sigma = (i^\sigma j^\sigma k^\sigma \dots)$.

$$\alpha(i_1 i_2 \dots i_k) \alpha^{-1} = (\alpha(i_1) \alpha(i_2) \dots \alpha(i_k)).$$

$$\alpha^t(i_1 i_2 \dots i_t) \alpha = (\alpha(i_1) \alpha(i_2) \dots \alpha(i_t))$$

$$\alpha^t(i_1 i_2 \dots i_t) \alpha \text{ group action.} \\ (\alpha(i_j)) = (i_j)^{(i_1 \dots i_t) \alpha} = (i_{j+1}) \alpha = \alpha(i_{j+1})$$

Remark 6. 1. $|\text{Fix}(g)| = n - 3 \Leftrightarrow g$ is a 3-cycle. ✓.

2. As g is even, if g is not a 3-cycle, then either g moves at least 5 points or g is a product of two disjoint transpositions. ✓.

Claim 2: N contains all 3-cycles.

Remark 7. each 3-cycle is conjugate to (123) by even permutations. Let (ijk) be an arbitrary 3-cycle.

Then there exists $x \in S_n$ such that $(ijk)^x = (123)$.

1. If $\{i, j, k\} \cap \{1, 2, 3\} = \emptyset$, then let $x = (1i)(2k)(3j)(23)$. Then $(ijk)^x = (123)$.

or. 2. Assume $|\{i, j, k\} \cap \{1, 2, 3\}| = 1$. WLOG, $i = 1$ and $\{j, k\} \cap \{2, 3\} = \emptyset$. Let $x = (2j)(3k)$.

Then $(ijk)^x = (123)$.

3. Assume $|\{i, j, k\} \cap \{1, 2, 3\}| = 2$. WLOG, $i = 1, j = 2$ and $k = 4$. Let $x = (k35)$.

That means all 3-cycles are in N , since $N \triangleleft A_n$, and each element of A_n is a product of 3-cycles so $N = A_n$.

Remark 8. If G is simple then for all $e \neq g \in G$, the normal closure $g^G = G$ where $g^G = \langle g^x | x \in G \rangle$.

Permutation Group

由双射组成的 group.

Definition 9. For a set Ω , each subgroup of $\text{Sym}(\Omega)$ is called a permutation group on Ω .

Theorem 10. (Cayley) Every finite group is isomorphic to a permutation group.

证明. Let G be a group, $G = \{g_1, g_2, \dots, g_n\} = \Omega$. For each element $x \in G$ define a permutation on Ω by $\hat{x} : g_i \mapsto g_i x$. Then $\hat{x} \in \text{Sym}(\Omega)$ (the proof of $\hat{x}$ is a bijection is easy to check).

Let $\hat{G} = \{\hat{x} | x \in G\}$ then $\hat{G}$ is a group which is isomorphic to G since $\hat{x}\hat{y} = \widehat{xy}$. Also $\hat{G} \leq \text{Sym}(\Omega)$ □

For any $g_i, g_j \in G$ there exists $\hat{x} \in \hat{G}$ s.t. $g_i \hat{x} = g_j$. So $\hat{G}$ is transitive on Ω . And such $\hat{x}$ is unique. So $\hat{G}$ is regular on Ω which means only the identity permutation fixes some point of Ω .

下面证明 H 含有所有的 3 轮换. 不妨设上述的 $\tau = (123)$.

对于任一 $(ijk) \in A_n$, 取

$$\nu = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & \dots & n \\ i & j & k & x_4 & x_5 & \dots & x_n \end{pmatrix},$$

其中 $(x_4 x_5 \dots x_n)$ 是 $\{1, 2, \dots, n\} \setminus \{i, j, k\}$ 的一个适当的排列, 使得 $\nu \in A_n$. 这样适当的排列是可以取到的, 原因是: 如果 ν 是奇置换, 将 ν 的表达式中的 x_4 和 x_5 对调, 即令

$$\nu' = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & \dots & n \\ i & j & k & x_5 & x_4 & \dots & x_n \end{pmatrix},$$

则 $\nu' = (x_4 x_5)\nu$ 是偶置换. 由引理 1.10 即知 $(ijk) = \nu\tau\nu^{-1} \in H$. 这就完成了定理的证明. □